

PROJECT DOCUMENTATION

Global Sales Performance Dashboard

Name : ADITYA KUMAR

Roll No : 2306089

Batch : 2023-27

Program : BACHELOR'S IN INFORMATION
TECHNOLOGY

1. Title & Problem Statement

1.1 Project Title

Global Sales Performance Dashboard — A comprehensive, interactive Power BI dashboard built to visualize and analyze global retail sales data across regions, product categories, and customer segments.

1.2 Problem Statement

In modern retail environments, businesses generate enormous volumes of transactional data daily. However, decision-makers often lack an efficient, centralized view of sales performance, profit trends, and regional contributions. This results in:

- Delayed identification of underperforming regions or product categories
- Inability to track profit margins and key metrics at a glance
- Difficulty in understanding the relationship between sales volume and profitability
- Lack of interactive, filterable views for executives and analysts

This project addresses these challenges by building a multi-visual, interactive Power BI dashboard that transforms raw SuperStore sales data into actionable business intelligence.

2. Solution & Features

2.1 Solution Overview

The Global Sales Performance Dashboard is a single-page, fully interactive Power BI report that consolidates all critical sales KPIs and visual analytics into one unified view. The solution covers data ingestion, transformation, modeling, and visualization — following industry best practices.

2.2 Dashboard KPIs

13M Total Sales	1M Total Profit	51.3K Total Orders	11.6% Profit Margin
---------------------------	---------------------------	------------------------------	-------------------------------

These KPI cards are positioned at the top of the dashboard, providing instant visibility into the four most critical business metrics before diving into detailed analysis.

2.3 Key Features & Visuals

- KPI Cards — Total Sales, Total Profit, Total Orders, and Profit Margin displayed as headline metrics
- Scatter Plot (Sales vs Profit by Category) — Reveals profitability relationships across Furniture, Office Supplies, and Technology with color-coded dots per category
- Region-wise Sales & Profit Matrix Table — Tabular breakdown of all 13 global regions, showing Total Sales and Total Profit side by side with expandable rows
- Geographical Map (Sales Distribution by Country) — Interactive bubble map showing sales concentration globally across North America, Europe, Asia, Africa, and Oceania
- Region Slicer — Dropdown filter enabling users to drill into any region (Africa, Canada, Caribbean, Central, Central Asia, East, EMEA, etc.)
- Dynamic Cross-Filtering — All visuals respond to slicer and chart selections in real time

3. Tech Stack

The following technologies were used to build and deploy this project end-to-end:

Component	Technology Used
Visualization Tool	Microsoft Power BI Desktop
Data Source	SuperStore Orders CSV (Kaggle)
Data Transformation	Power Query Editor
Calculations & Metrics	DAX (Data Analysis Expressions)
Data Model	Star Schema (Fact + Dimension Tables)
Publishing	Power BI Service (Cloud)
Version Control	GitHub (.pbix, README, CSV)

3.1 DAX Measures Used

- Total Sales = SUM(Orders[Sales])
- Total Profit = SUM(Orders[Profit])
- Total Orders = COUNTROWS(Orders)
- Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
- Region-wise Aggregations using CALCULATE() with filter context

4. Dashboard Screenshots

The following section describes the key visual panels visible in the dashboard preview hosted on GitHub (github.com/adityakumar2005jsr-sketch/PowerBI-Sales-Dashboard).

4.1 Main Dashboard View

The main dashboard view displays all visuals in a single scrollable canvas. The header reads 'Global Sales Performance Dashboard' in a dark navy title bar, followed by four KPI cards (Total Sales: 13M, Total Profit: 1M, Total Orders: 51.3K, Profit Margin: 11.6%).

4.2 Scatter Plot — Sales vs Profit by Category

A scatter plot positioned in the bottom-left quadrant plots individual sub-category data points across three product categories (Furniture, Office Supplies, Technology). The X-axis represents Total Sales (0 to 100K) and the Y-axis represents Total Profit (-10K to 30K), revealing that Technology products tend to generate higher profit per sale, while some Furniture sub-categories show negative profit margins.

4.3 Region-wise Matrix Table

The center panel contains a detailed matrix table listing 13 regions. Central leads with 2,822,399 in Total Sales and 311,403.98 in Total Profit. The Total row at the bottom aggregates to 12,642,905 in Sales and 1,469,034.82 in Total Profit, confirming the KPI card values.

4.4 Geographical Sales Map

The right panel displays a world bubble map where circle size indicates sales volume by country. North America and Europe show the highest concentration of large bubbles, while Asia and Australia show moderate activity, and Africa/Caribbean regions display smaller, emerging market footprints.

5. Unique Points

This project differentiates itself from standard Power BI practice dashboards in the following ways:

- **Global Scope** — Unlike most practice projects that focus on a single country (typically the US), this dashboard uses a global dataset spanning 13 regions across all continents, adding complexity to the geographical and regional analysis.
- **Multi-Visual Integration** — The dashboard combines five distinct visual types (KPI Cards, Scatter Plot, Matrix Table, Map, and Slicer) in a single cohesive layout, demonstrating proficiency in visual storytelling and layout design.
- **Profitability-Focused Analysis** — The scatter plot reveals non-obvious insights such as Furniture sub-categories operating at a loss, which is not apparent from top-line sales figures alone.
- **GitHub Portfolio Deployment** — The project is publicly hosted on GitHub with a well-structured README, dashboard preview image, and the original .pbix and .csv files, making it accessible for portfolio review by recruiters.
- **Interactive Region Filtering** — The region slicer dynamically filters all visuals simultaneously, enabling stakeholders to perform rapid region-specific analysis without navigating multiple report pages.
- **Clean Dark-Header Design** — The professional navy header and consistent color palette adhere to enterprise dashboard design standards, distinguishing it from generic practice projects.

6. Future Improvements

The following enhancements are planned for subsequent iterations of this dashboard:

- **Time Intelligence & Trend Analysis** — Add a Line Chart showing Monthly Revenue Trend with YTD, QTD, and MoM growth calculations using DAX time intelligence functions (TOTALYTD, PREVIOUSMONTH, DATESINPERIOD).
- **Sales vs Target Waterfall Chart** — Integrate a target dataset to build an Actual vs Target Waterfall chart, enabling variance analysis and highlighting over/under-performance by month and region.
- **Customer Segmentation Page** — Add a dedicated page with a Pie Chart and Bar Chart for customer segment analysis (Consumer, Corporate, Home Office) to identify high-value segments.
- **Product Category Drill-Through** — Implement drill-through functionality allowing users to click on a category in the scatter plot and navigate to a sub-category-level detail page.
- **Row-Level Security (RLS)** — Configure RLS roles so that regional managers can only see data for their assigned regions when the report is shared via Power BI Service.
- **Mobile Layout Optimization** — Create a dedicated mobile view layout using Power BI's mobile formatting feature, ensuring the dashboard is accessible and readable on smartphones and tablets.
- **AI-Powered Insights** — Integrate Power BI's built-in AI visuals (Key Influencers, Decomposition Tree) to automatically surface the top drivers of profit margin and sales growth.
- **Automated Data Refresh** — Connect the dataset to a cloud-based data source (e.g., SharePoint or Azure SQL) and configure scheduled refresh in Power BI Service for real-time or daily updates.

End of Project Documentation

Global Sales Performance Dashboard | Power BI | April 2026